



Functional Description

Gas Genset

MTU Interface description

MIP4000 from V3.5 onward

Cabling and software communication

MS61019/02E

Name	SLU name, model	Engine model	Application group
MTU 8V4000 GS	GG08V4000A1 GG08V4000A2	8V4000L32	3A, continuous operation, unrestricted
MTU 8V4000 GS	GG08V4000A1 GG08V4000A2	8V4000L33 reduced power	3A, continuous operation, unrestricted
MTU 8V4000 GS	GG08V4000A1 GG08V4000A2	8V4000L32ER	3A, continuous operation, unrestricted
MTU 8V4000 GS	GG08V4000A1	8V4000L64	3A, continuous operation, unrestricted
MTU 12V4000 GS	GG12V4000A1 GG12V4000A2	12V4000L32	3A, continuous operation, unrestricted
MTU 12V4000 GS	GG12V4000A1 GG12V4000A2	12V4000L33 reduced power	3A, continuous operation, unrestricted
MTU 12V4000 GS	GG12V4000A1 GG12V4000A2	12V4000L32ER	3A, continuous operation, unrestricted
MTU 12V4000 GS	GG12V4000A1	12V4000L64	3A, continuous operation, unrestricted
MTU 16V4000 GS	GG16V4000A1 GG16V4000A2	16V4000L32	3A, continuous operation, unrestricted
MTU 16V4000 GS	GG16V4000A1 GG16V4000A2	16V4000L33 reduced power	3A, continuous operation, unrestricted
MTU 16V4000 GS	GG16V4000A1 GG16V4000A2	16V4000L32ER	3A, continuous operation, unrestricted
MTU 16V4000 GS	GG16V4000A1	16V4000L64	3A, continuous operation, unrestricted
MTU 20V4000 GS	GG20V4000A1 GG20V4000A2	20V4000L32	3A, continuous operation, unrestricted
MTU 20V4000 GS	GG20V4000A1 GG20V4000A2	20V4000L33 reduced power	3A, continuous operation, unrestricted
MTU 20V4000 GS	GG20V4000A1 GG20V4000A2	20V4000L32ER	3A, continuous operation, unrestricted
MTU 20V4000 GS	GG20V4000A1	20V4000L64	3A, continuous operation, unrestricted

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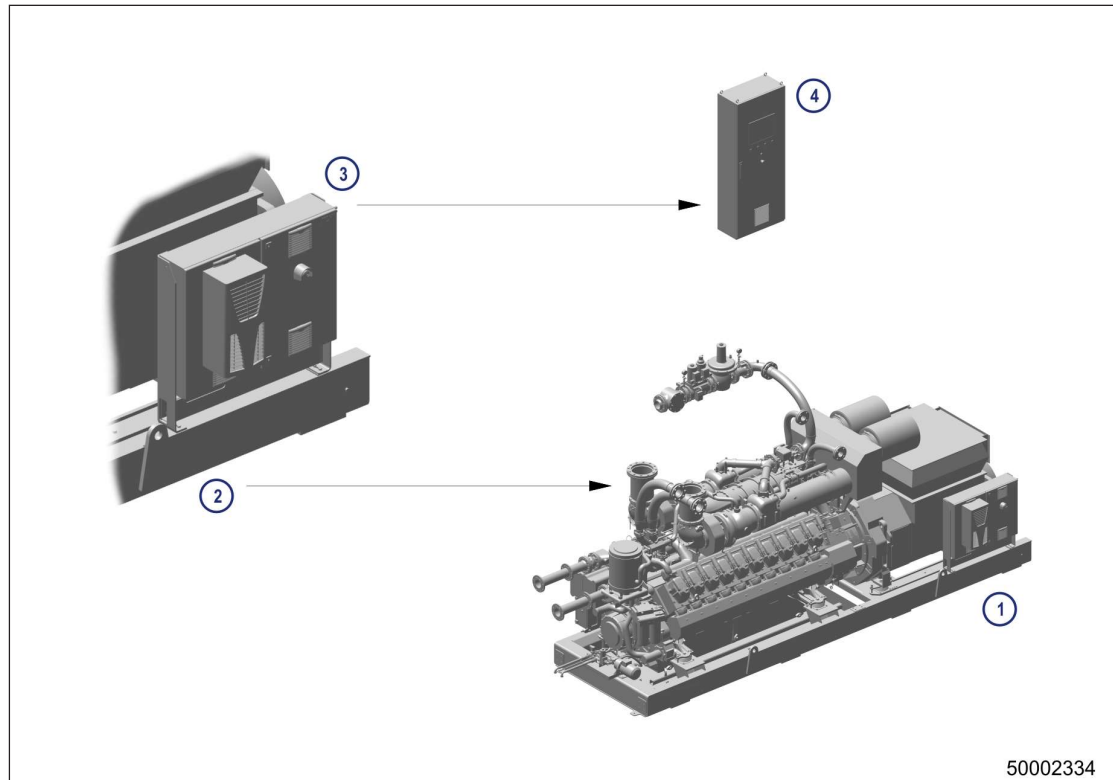
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1 Interface Description

1.1 General information

MTU Onsite Energy is responsible for interface clarification between the genset and customer system controller. All the interface signals between the genset and the customer system controller are operated from the MIP4000.



- | | |
|----------------------|------------------------------|
| 1 Genset | 3 Customer interface |
| 2 Engine connections | 4 Customer system controller |

1.2 Scope of services

- Communication with the engine governor via hardware signals and CAN.
- Communication with the customer system via hardware signals and Ethernet.
- Control elements (EMERGENCY STOP buttons, main switch auxiliary drives).
- Integrated safety chains for engine EMERGENCY STOP and manual EMERGENCY STOP.
- Generator current measuring transformer wired in the scope of supply.
- Generator measurement voltage at low-voltage wired in the scope of supply.

Generator protection

- Overfrequency (ANSI: 81, IEC: f>)
- Underfrequency (ANSI: 81, IEC: f<)
- Overvoltage (ANSI: 59, IEC: U>)
- Undervoltage (ANSI: 27, IEC: U<)
- Normally dependent overcurrent (ANSI: 51, IEC: I>)
- Overcurrent (ANSI: 50, IEC: I>>)
- Overload (ANSI: 32F, IEC: Pf>>)
- Reverse power (ANSI: 32R, IEC: Pr>)
- Unbalanced load (ANSI: 46, IEC: I2>)

Generator monitoring -/ measurement

- Generator effective power - P (kW)
- Generator reactive power - Q (kVAr)
- Generator apparent power - S (kVA)
- Generator frequency
- Generator voltage U12, U23, U31
- Generator current L1, L2, L3
- Generator cos phi
- Generator active energy (kWh)

Busbar monitoring -/ measurement

- Busbar voltage U12, U23, U31
- Busbar frequency

Mains monitoring / measurement

- Mains voltage U12, U23, U31
- Mains frequency

Synchronization

- Generator circuit breaker (GCB):
Only floating contacts, power section in the customer system.
- Mains circuit breaker (MCB):
Only floating contacts, power section in the customer system.

Generator temperature monitoring

- Winding temperature monitoring U1/V1/W1 and limit value monitoring (including wiring in the scope of supply)
- Bearing temperature monitoring driving/driven side and limit value monitoring (including wiring in the scope of supply)

Generator functions

- Activation of the generator standstill heating (power element including wiring in the scope of supply)
- Activation of the voltage regulator (including wiring in the scope of supply)
- Activation of de-excitation of generator (power element including wiring in the scope of supply)

Gas system (monitoring and safety function)

- Gas pressure min. / max. (safety function)
- Activation of the gas leak test (power element in the scope of supply)
- Activation of the gas solenoid valves (power elements in the scope of supply)

General controller functions

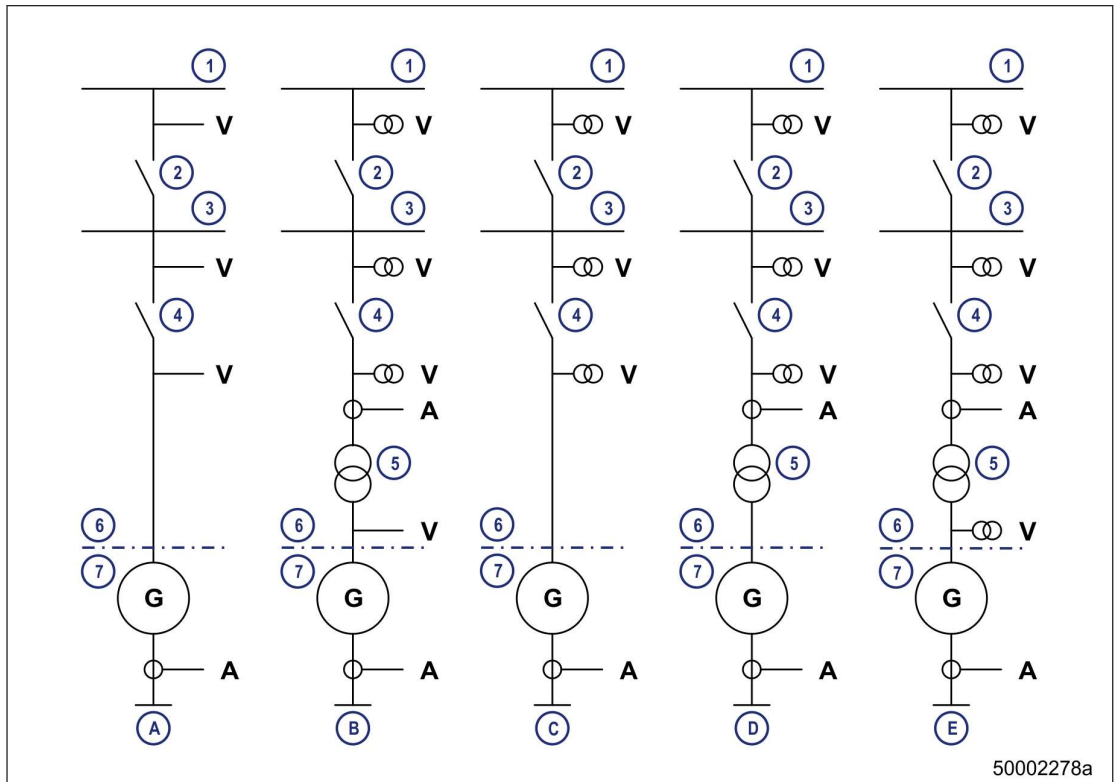
- Activation of the auxiliary drives (potential-free contact or software signal)
- Open activation of backup-switch (potential-free contact or software signal)
- Activation lubricating oil solenoid valve (power element including wiring in the scope of supply)
- Activation of the used oil / oil priming pump (power element including wiring in the scope of supply)
- Activation used oil solenoid valve (power element including wiring in the scope of supply)
- Activation of the heating cartridges of the engine cooling water pre-heating (power elements including wiring in the scope of supply)

Electrical integration

The following variants are available for the electrical integration. For the variants with medium voltage step up transformer (Figure B / D / E), an MIP with a second energy measurement module is necessary.

Item	Designation
A	Low-voltage generator
B	Low-voltage generator coupled with medium-voltage step-up transformer
C	Medium-voltage generator
D	Medium-voltage generator linked with medium-voltage step-up transformer; no voltage transformer between medium-voltage generator and medium-voltage step-up transformer
E	Medium-voltage generator linked with medium-voltage step-up transformer; with voltage transformer between medium-voltage generator and medium-voltage step-up transformer

Table 1: Variants



Item	Designation
01	Mains
02	MCB (mains circuit breaker)
03	Busbar
04	GCB (generator circuit breaker)
05	Step-up transformer
06	Customer
07	MTU Oniste Energy

Table 2: Position designation

1.3 Requirements for the customer system

- Providing the 400 VAC auxiliary drive supply
- Providing the 24 VDC voltage supply, controller
- Providing the 24 VDC voltage supply, starter
- Providing the measurement voltages for the mains and busbars 100 VAC / 400 VAC

Only for medium voltage generator

Providing the measurement voltage for generator 100 VAC, see the chapter Electrical Integration (Figure C / E)

Only for medium voltage: Step-up transformer

- Providing the measurement voltage 100 VAC between the medium voltage step-up transformer and generator circuit breaker (GCB), see the chapter Electrical Integration (Figure B / D / E)
- Providing the measurement current 1 A between the medium voltage step-up transformer and generator circuit breaker (GCB), see the chapter Electrical Integration (Figure B / D / E)

Integration of a mains protection device with the monitors

- Mains over-frequency / Mains under-frequency
- Mains overvoltage / Mains undervoltage
- Vector shift
- Asymmetry
- Phase shift monitoring

Optional integration of a differential protection device

Optional integration of a ground fault protection device

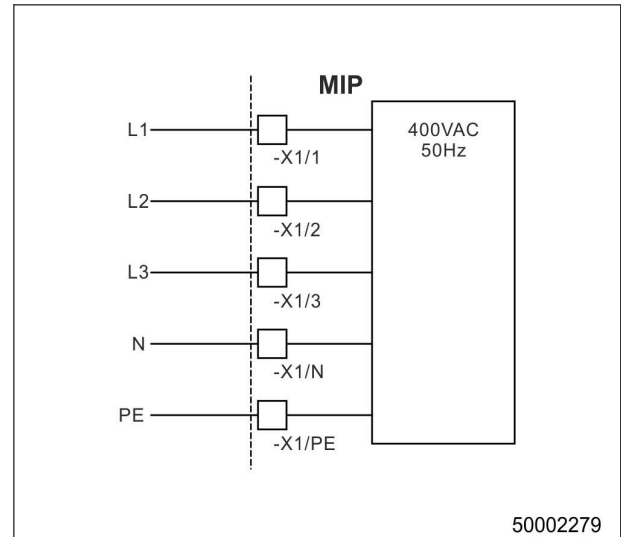
Provision of the power sections for

- External auxiliary drives
- Fresh oil pump
- Generator circuit breaker (GCB)
- Mains circuit breaker (MCB)

1.4 Hardware Signals

1.4.1 Hardware signals from outside

Power supply for auxiliary drives

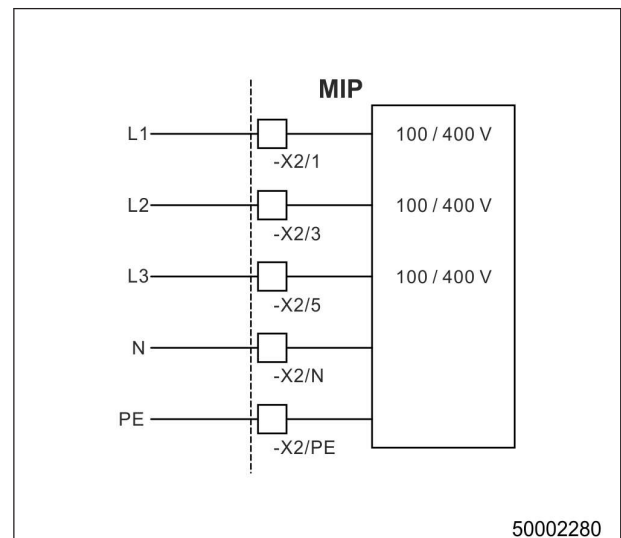


Maximum pre-fuse 25 A gG, maximum conditional rated short-circuit current I_{cc} 10 kA, maximum core cross-section 6 mm².

The following auxiliary drives are supplied power from this supply:

- Switch cabinet ventilation (optionally a cooling unit)
- Generator standstill heating
- Engine coolant preheating
- Used oil/oil priming pump

Mains measurement voltage



Maximum pre-fuse 10 A, maximum conditional rated short-circuit current I_{cc} 1 kA, maximum core cross-section 6 mm².

The 400 VAC or 100 VAC version (medium voltage) can be used as the mains rated voltage.

Version 400 VAC:

Measuring range 0 to 480 VAC

Version 100 VAC:

Measuring range 0 to 110 VAC

If the measurement voltage is changed from 400 VAC to 100 VAC, the parameters in the MIP4000 must be adjusted.

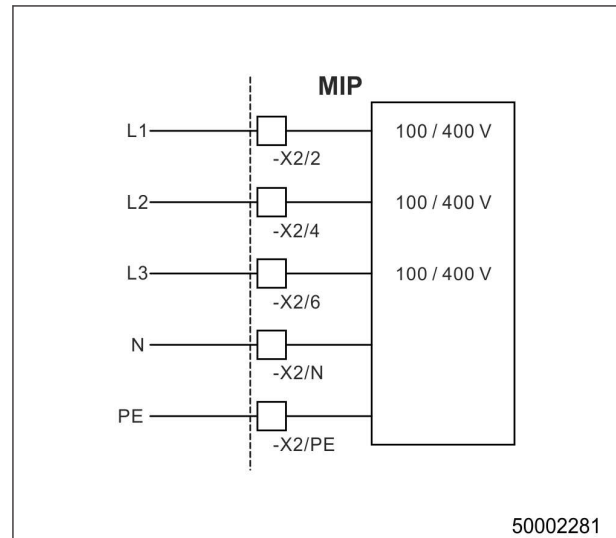
The mains measurement voltage is used for the mains monitoring/measurement and for synchronization of the mains circuit breaker (MCB).

Synchronization of MCB: Busbar voltage ↔ mains measurement voltage

For pure isolated operation (no public network available), this measurement voltage is not required.

With pure mains parallel operation without mains failure operation, this measurement voltage should be bridged with the measurement voltage of the busbar voltage, see the circuit diagram.

Busbar voltage



Maximum pre-fuse 10 A, maximum conditional rated short-circuit current I_{cc} 1 kA, maximum core cross-section 6 mm².

The 400 VAC or 100 VAC version (medium voltage) can be used as the busbar voltage.

Version 400 VAC:

Measuring range 0 to 480 VAC

Version 100 VAC:

Measuring range 0 to 110 VAC

If the measurement voltage is changed from 400 VAC to 100 VAC, the parameters in the MIP4000 must be adjusted.

The busbar voltage is used to synchronize the generator circuit breaker (GCB) and for the mains circuit breaker (MCB).

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